# AI Basics

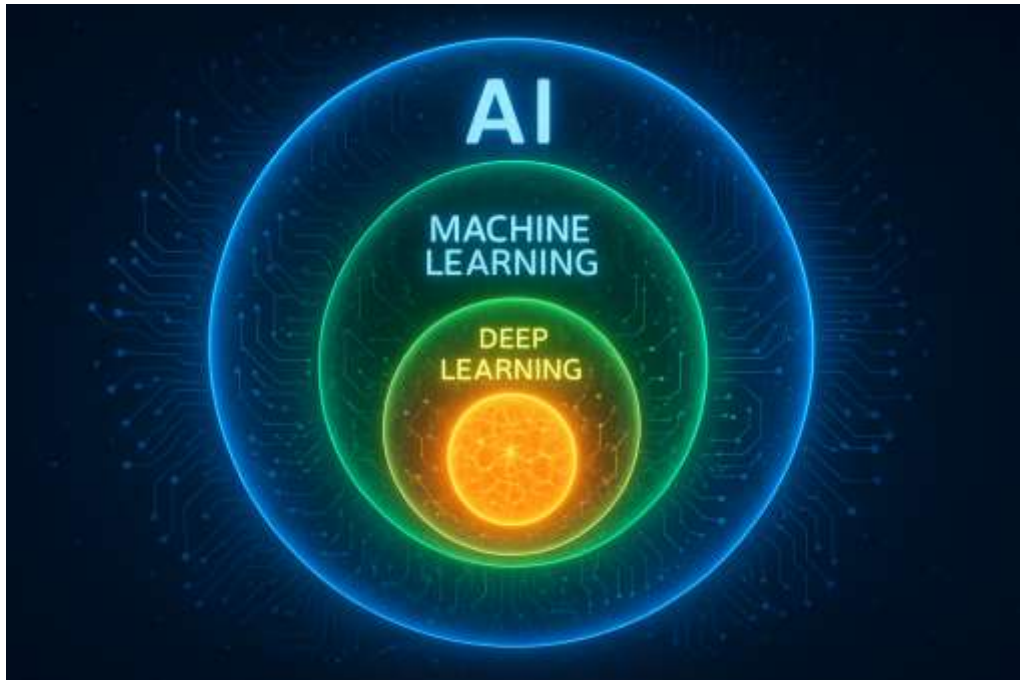
# What is Artificial Intelligence

**Artificial Intelligence (AI)** is the simulation of human intelligence processes by computer systems and machines. These systems learn from data, recognize patterns, reason, and make decisions to solve complex problems.

**Definition:** AI is an umbrella term for technology that allows machines to perform tasks typically requiring human cognitive functions—such as speech recognition, visual interpretation, and problem-solving.

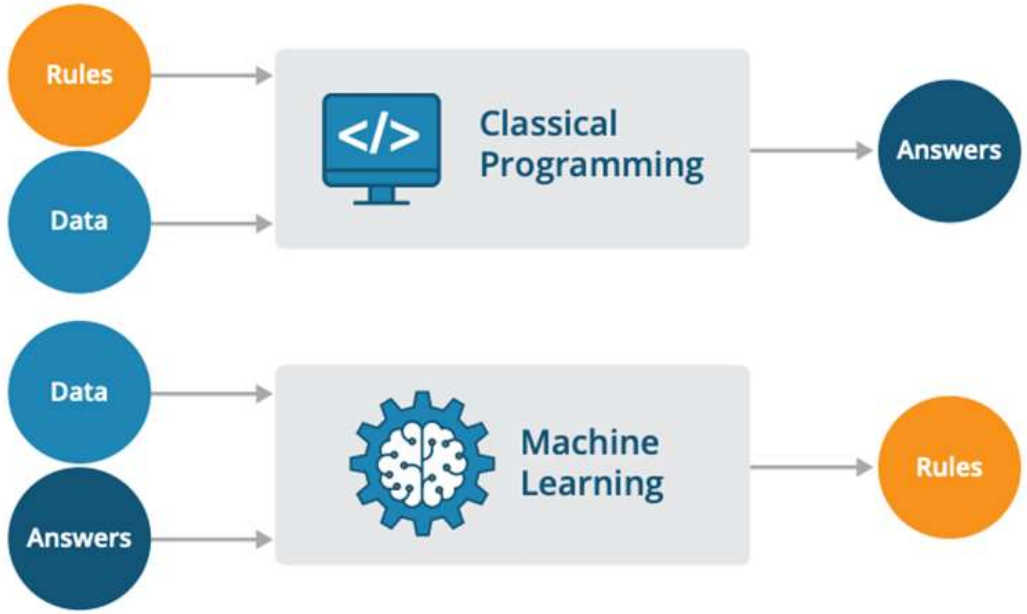
# Subsets of AI

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) form a nested hierarchy where each concept is a specialized subset of the one before it: AI is the largest umbrella, ML is a subset of AI, and Deep Learning is a subset of ML.



# Machine Learning

**Machine Learning** is a discipline in computer science where we train machines on data so that they can make predictions without explicit programming.



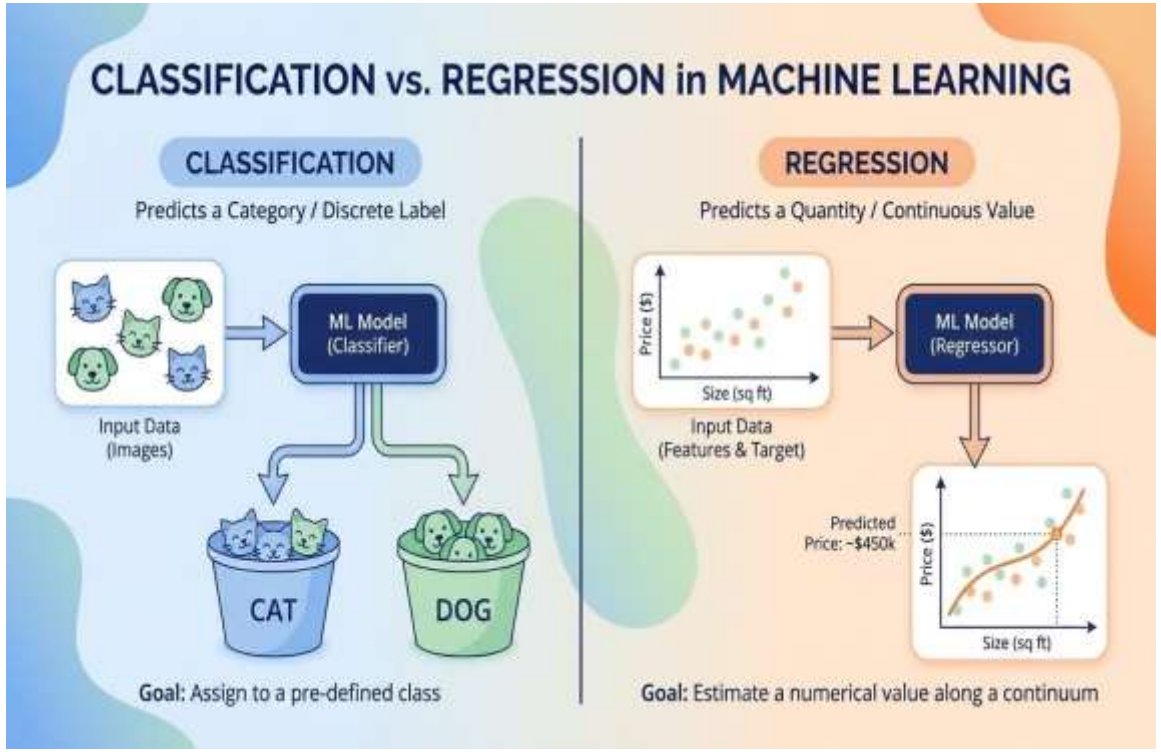
# Major Categories

**Classification:**

Classification assigns input data into distinct, predefined classes or labels.

**Regression:**

Regression models the relationship between independent variables and a continuous numerical target to estimate quantities.

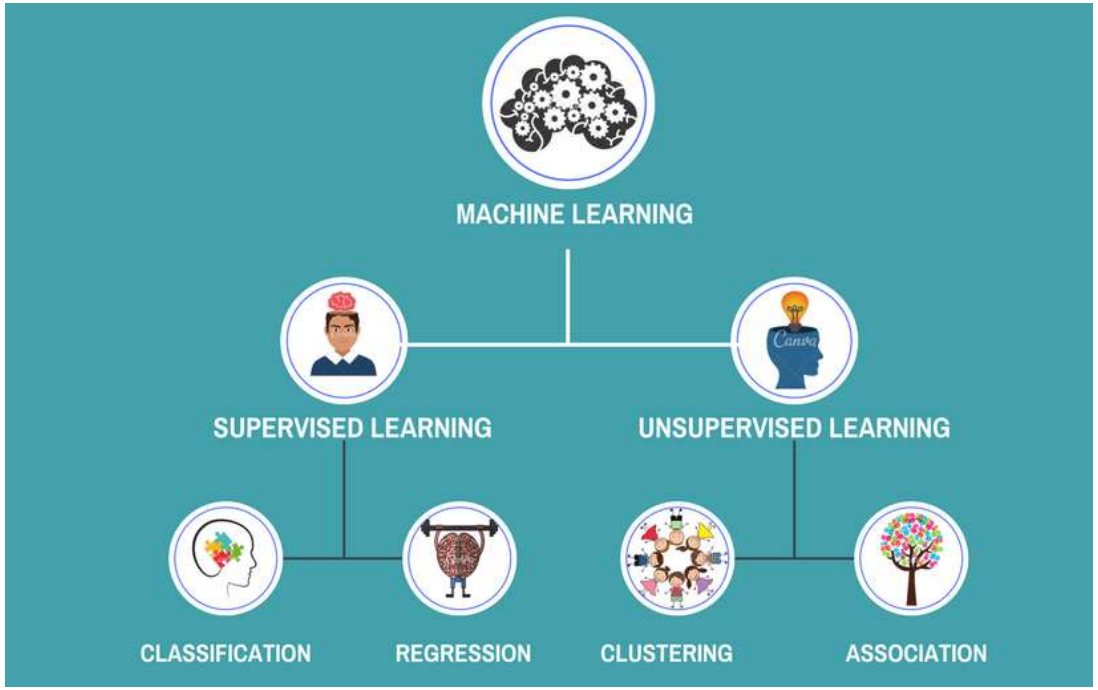


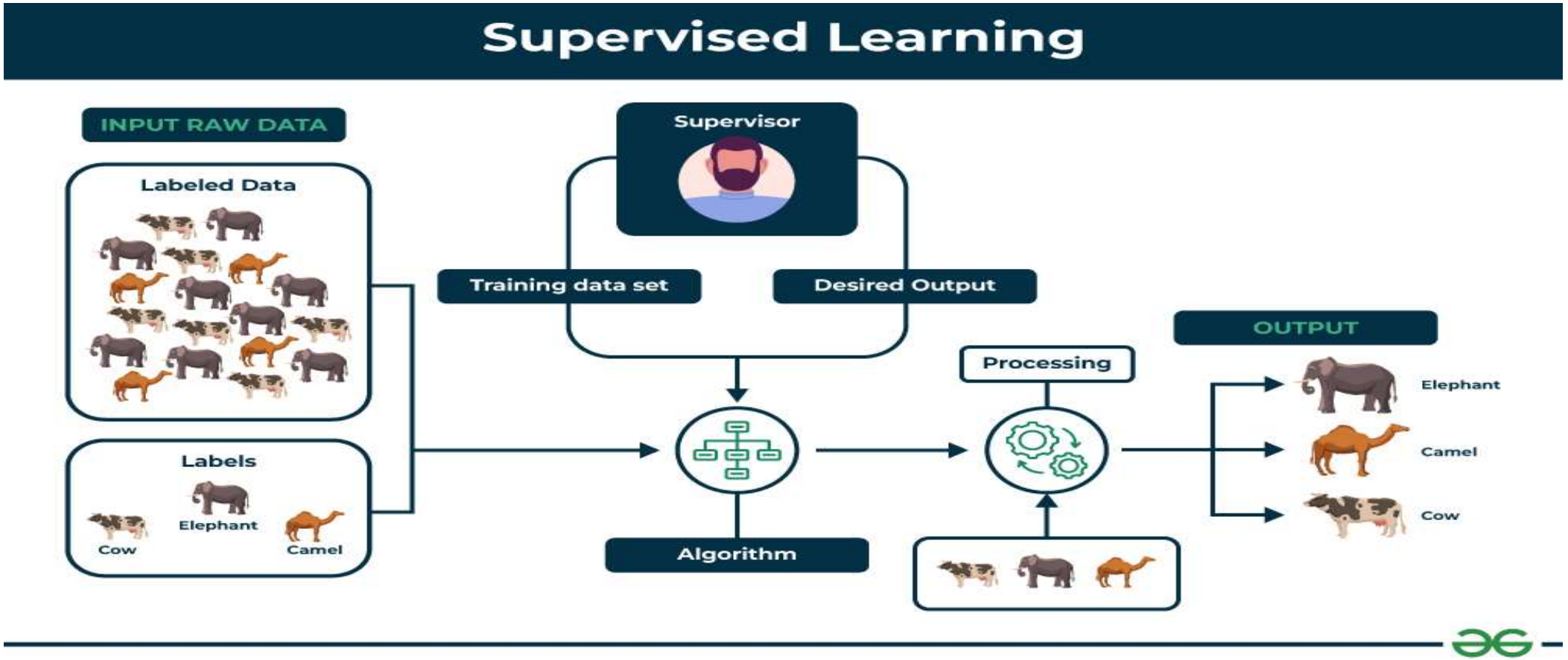
# Real World Examples

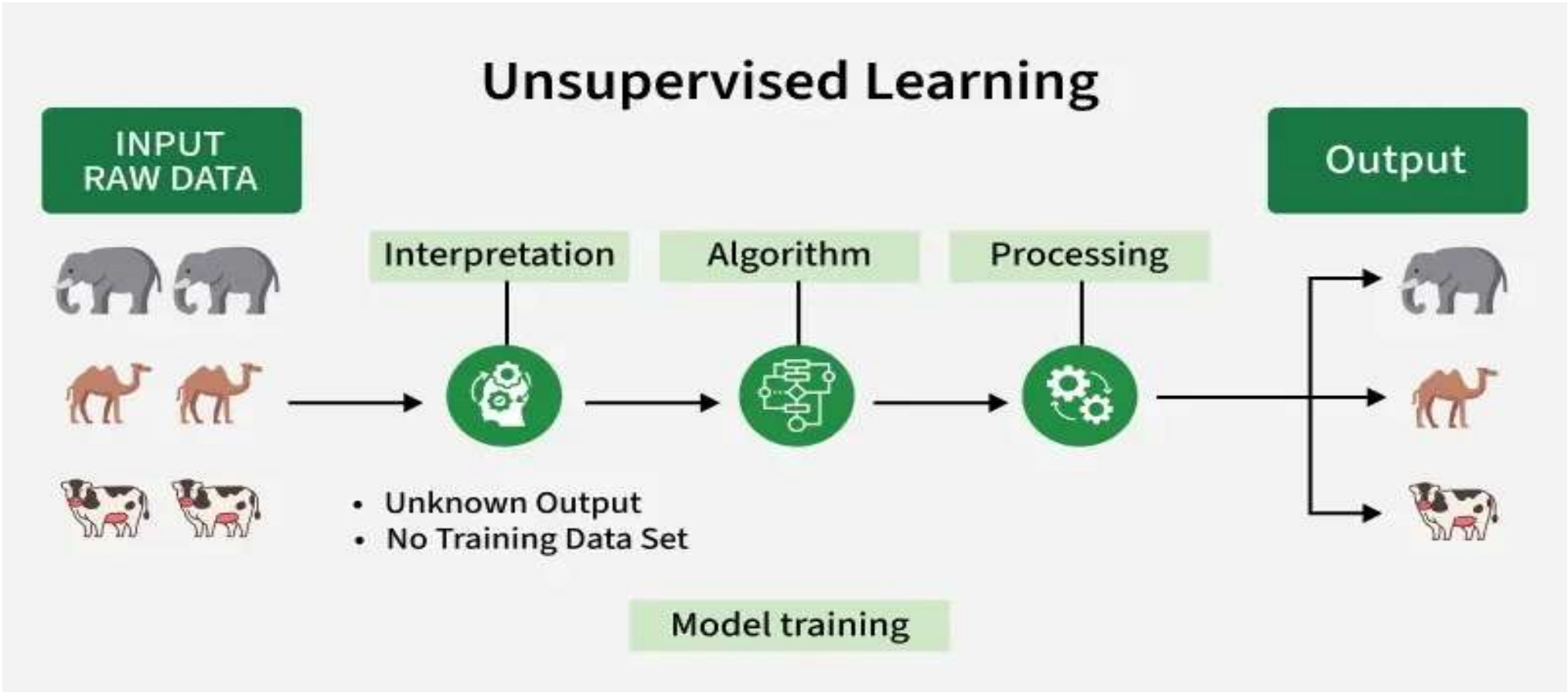
| Classification | Regression |
| --- | --- |
| Email SPAM Classification | Property Price Prediction |
| Fraud Detection | Salary Prediction |
| Image Classification | Stock Price Forecasting |

# Supervised and Unsupervised Learning

Supervised learning acts like a student learning with a teacher guiding them through a labeled answer key, while unsupervised learning acts like an explorer finding hidden structures in data completely on its own.

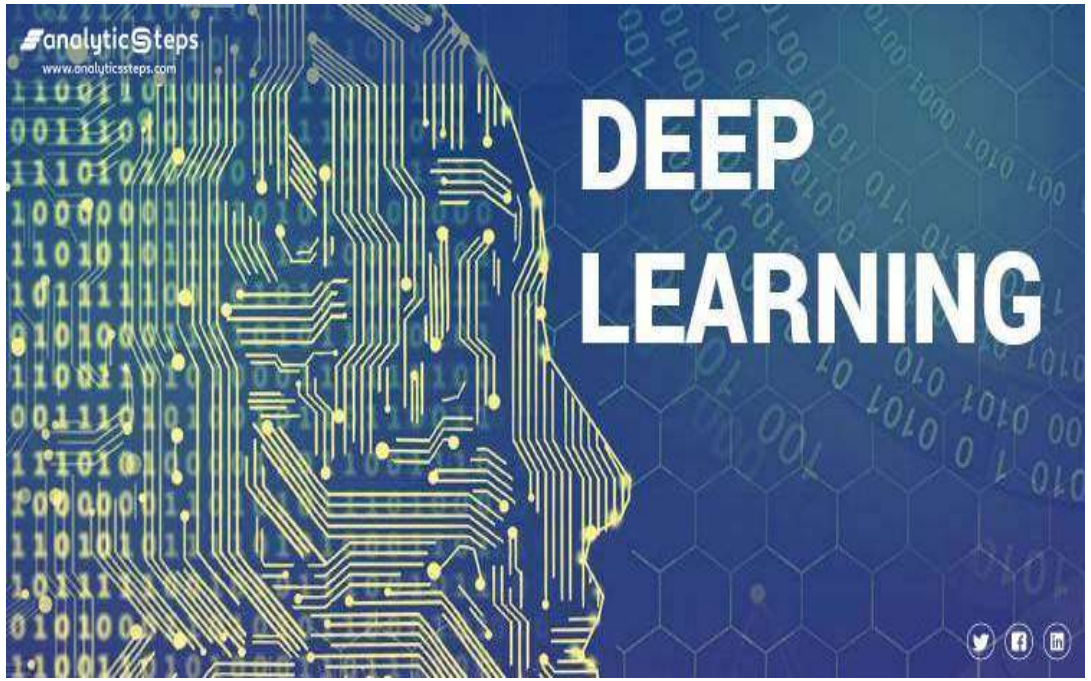



Supervised Learning
INPUT RAW DATA
Labeled Data
Labels
Cow
Elephant
Camel
Supervisor
Training data set
Desired Output
Algorithm
Processing
OUTPUT
Elephant
Camel
Cow


Unsupervised Learning
INPUT
RAW DATA
Interpretation
Algorithm
Processing
Output
• Unknown Output
• No Training Data Set
Model training

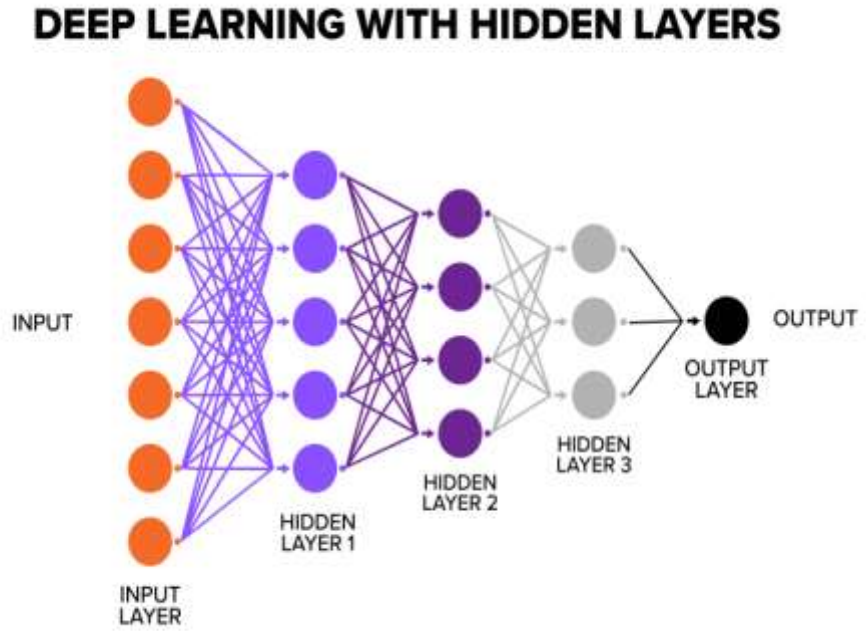
# Deep Learning

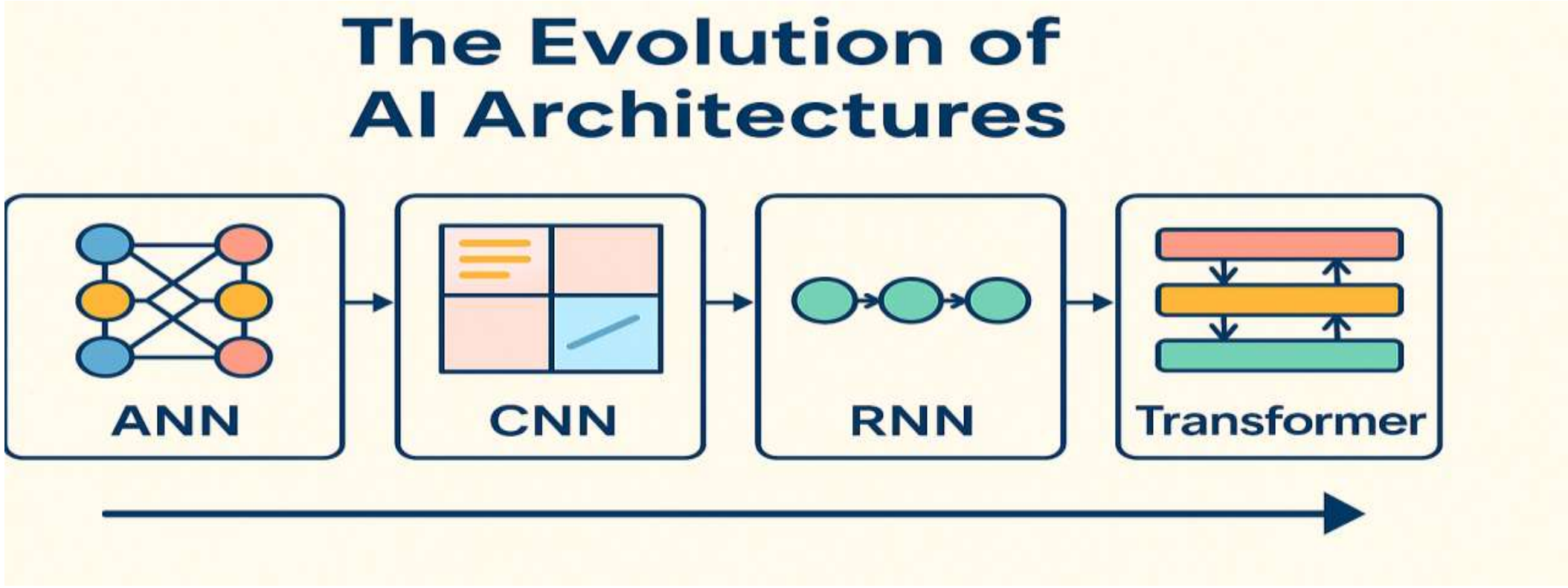
**Deep Learning** is a machine learning technique that uses neural networks to learn from large amounts of data, mimicking the human brain's ability to recognize patterns and make decisions.



# Neural Network

A **neural network** is a machine learning model inspired by the human brain, while deep learning is a subset of machine learning that uses multi-layered neural networks (typically with multiple hidden layers) to automatically learn complex features from data




The Evolution of
AI Architectures
ANN
CNN
RNN
Transformer

# Thank You